Game Theory

Midterm Examination: Solutions and Grading Rubric | Fall 2023

Prof. R. Alvarez

Grading notes for GSIs

- Total 60 points. Point values for each step are listed in the rubric boxes; award partial credit only as described there.
- Carry-forward rule: if a student makes an arithmetic error early in a part and then proceeds correctly with the wrong number, deduct only once (at the step where the error occurs).
- Equivalent descriptions (e.g., probabilities written as decimals that are exactly correct) receive full credit.

Problem 1 Iterated Dominance [12 points]

	a	b	c	d
U	4, 0	1, 3	2, 1	0, 2
M	0, 4	3, 2	2, 1	0, 1
D	1, 1	2, 3	1, 5	4, 2

(a) Column's payoffs from b are $(3, 2, 3)$ against (U, M, D) and from d are $(2, 1, 2)$. Since $3 > 2$, $2 > 1$, $3 > 2$, d **is strictly dominated by** b. No other pure strategy is dominated by a pure strategy, because each of the others is the *unique* best response to some opponent strategy: U to a (4), M to b (3), D to d (4); a to M (4), b to U (3), c to D (5).

Rubric

1 pt: identifies d dominated by b with the three comparisons. 1 pt: argues that no other pure strategy is dominated by a pure strategy (checking pairs or the unique-best-response argument). A bare claim with no check earns 0 for this point.

(b) With d removed, Row's payoffs over (a, b, c) are $U = (4, 1, 2)$, $M = (0, 3, 2)$, $D = (1, 2, 1)$. U does not dominate D ($1 < 2$ against b), and M does not dominate D ($0 < 1$ against a). For the mixture,

$$\alpha U + (1-\alpha)M = \big(4\alpha,\ 3-2\alpha,\ 2\big) > (1,\ 2,\ 1) \iff 4\alpha > 1 \text{ and } 3-2\alpha > 2 \iff \tfrac{1}{4} < \alpha < \tfrac{1}{2}.$$

So D is strictly dominated by $\alpha U + (1-\alpha)M$ exactly for $\alpha \in (\frac{1}{4}, \frac{1}{2})$; e.g. $\alpha = \frac{1}{3}$ gives $(\frac{4}{3}, \frac{7}{3}, 2)$. (Before d is removed no such α exists: the mixture earns $0 < 4$ against d.)

Rubric

1 pt: shows neither U nor M alone dominates D. 1 pt: writes the mixture's payoff vector correctly. 2 pts: correct open interval $(\frac{1}{4}, \frac{1}{2})$. Give 1 of these 2 pts for a single valid α with verification but no interval, or for a closed interval $[\frac{1}{4}, \frac{1}{2}]$ (endpoints give only weak dominance).

(c) Without D, Column's payoffs over (U, M) are $a = (0, 4)$, $b = (3, 2)$, $c = (1, 1)$; now c is strictly dominated by b. In the remaining $\{U, M\} \times \{a, b\}$ game, each strategy is the unique best response to one of the opponent's strategies, so nothing else is eliminated. The surviving game is

	a	b
U	4, 0	1, 3
M	0, 4	3, 2

Rubric

1 pt: eliminates c (dominated by b) after D is gone. 1 pt: correct 2×2 reduced game with a reason why elimination stops.

(d) Strictly dominated strategies are never played in a Nash equilibrium, and IESDS does not change the set of equilibria, so it suffices to solve the reduced game. It has no pure equilibrium (best responses cycle: $(U,a) \to (U,b) \to (M,b) \to (M,a) \to (U,a)$). Let $p = \Pr(U)$, $q = \Pr(a)$.

- Column indifferent: a gives $4(1-p)$, b gives $3p + 2(1-p) = 2 + p$. So $4 - 4p = 2 + p$, i.e. $p = \frac{2}{5}$.
- Row indifferent: U gives $4q + (1-q) = 1 + 3q$, M gives $3(1-q)$. So $1 + 3q = 3 - 3q$, i.e. $q = \frac{1}{3}$.

The unique Nash equilibrium of the original game is

$$\text{Row: } \left(\tfrac{2}{5}, \tfrac{3}{5}, 0\right) \text{ on } (U, M, D), \qquad \text{Column: } \left(\tfrac{1}{3}, \tfrac{2}{3}, 0, 0\right) \text{ on } (a, b, c, d),$$

with expected payoffs $\boxed{u_R = 2,\ u_C = \frac{12}{5}}$. Check against the eliminated strategies: D earns $\frac{1}{3} \cdot 1 + \frac{2}{3} \cdot 2 = \frac{5}{3} < 2$; c earns $1 < \frac{12}{5}$ and d earns $\frac{2}{5} \cdot 2 + \frac{3}{5} \cdot 1 = \frac{7}{5} < \frac{12}{5}$.

Rubric

1 pt: notes no pure NE. 1 pt: $p = \frac{2}{5}$. 1 pt: $q = \frac{1}{3}$. 1 pt: states the equilibrium in the *original* game (zeros on D, c, d) with both payoffs.
Common error: swapping which player's indifference determines which probability (p from Row's payoffs). Award at most 2/4 in that case.

Problem 2 Pure and Mixed Equilibria in a 2×3 Game [14 points]

	L	C	R
T	3, 4	1, 3	2, 0
B	1, 0	4, 2	4, 3

(a) Row's best responses: T to L; B to C; B to R. Column's best responses: L to T; R to B. Pure Nash equilibria: $\boxed{(T, L) \text{ and } (B, R)}$.

Rubric

1 pt: correct best responses (marked in the matrix is fine). 2 pts: both equilibria, and no others (1 pt each; -1 for each extra profile listed).

(b) With $p = \Pr(T)$:

$$u_C(L) = 4p, \qquad u_C(C) = 3p + 2(1-p) = 2 + p, \qquad u_C(R) = 3(1-p) = 3 - 3p.$$

$C = R$ at $p = \frac{1}{4}$; $L = C$ at $p = \frac{2}{3}$; $L = R$ at $p = \frac{3}{7}$, but there C gives $\frac{17}{7} > \frac{12}{7}$, so that crossing is irrelevant. Hence

$$B_C(p) = \begin{cases} R & 0 \le p < \frac{1}{4}, \\ \text{any mixture of } C, R & p = \frac{1}{4}, \\ C & \frac{1}{4} < p < \frac{2}{3}, \\ \text{any mixture of } L, C & p = \frac{2}{3}, \\ L & \frac{2}{3} < p \le 1. \end{cases}$$

Rubric

1 pt: the three payoff lines. 2 pts: the two relevant thresholds $\frac{1}{4}, \frac{2}{3}$ (1 each). 1 pt: complete correspondence including mixtures at the thresholds. Students who list $p = \frac{3}{7}$ as a switching point lose this last point.

(c) If Column plays a pure strategy, Row has a unique best response, so Row does not mix; and at $p \in \{0, 1\}$ Column's best response is unique, so Column does not mix either. Hence in a non-pure equilibrium Row mixes, Row must be indifferent, and Column must mix; by (b) this requires $p \in \{\frac{1}{4}, \frac{2}{3}\}$.

- $p = \frac{1}{4}$: Column mixes C (prob. r) and R. Row: $u_R(T) = r + 2(1 - r) = 2 - r$, $u_R(B) = 4$. Since $2 - r < 4$ for every r, Row strictly prefers B: no equilibrium.
- $p = \frac{2}{3}$: Column mixes L (prob. q) and C. Row: $u_R(T) = 3q + (1 - q) = 1 + 2q$, $u_R(B) = q + 4(1 - q) = 4 - 3q$. Indifference gives $q = \frac{3}{5}$.

The unique non-pure equilibrium is $\boxed{\text{Row } (\frac{2}{3}, \frac{1}{3}), \quad \text{Column } (\frac{3}{5}, \frac{2}{5}, 0)}$.

Rubric

1 pt: argues that Row must be indifferent and Column must mix at a threshold. 1 pt: rules out $p = \frac{1}{4}$. 2 pts: $q = \frac{3}{5}$ from Row's indifference. 1 pt: states the equilibrium with R unused.
A student who checks only the $\{L, C\}$ support and never addresses $\{C, R\}$ or $\{L, R\}$ earns at most 3/5.

(d) Payoffs: $(T, L) \mapsto (3, 4)$; $(B, R) \mapsto (4, 3)$; mixed $\mapsto \left(1 + 2 \cdot \frac{3}{5},\ 2 + \frac{2}{3}\right) = (\frac{11}{5}, \frac{8}{3})$. The mixed equilibrium is Pareto dominated by *both* pure equilibria; the two pure equilibria are not Pareto ranked.

Rubric

1 pt: the three payoff pairs. 1 pt: correct Pareto comparison.

Problem 3 A Zero-Sum Game: the Graphical Method [16 points]

	c_1	c_2	c_3	c_4
r_1	5	-2	2	0
r_2	-1	4	1	6

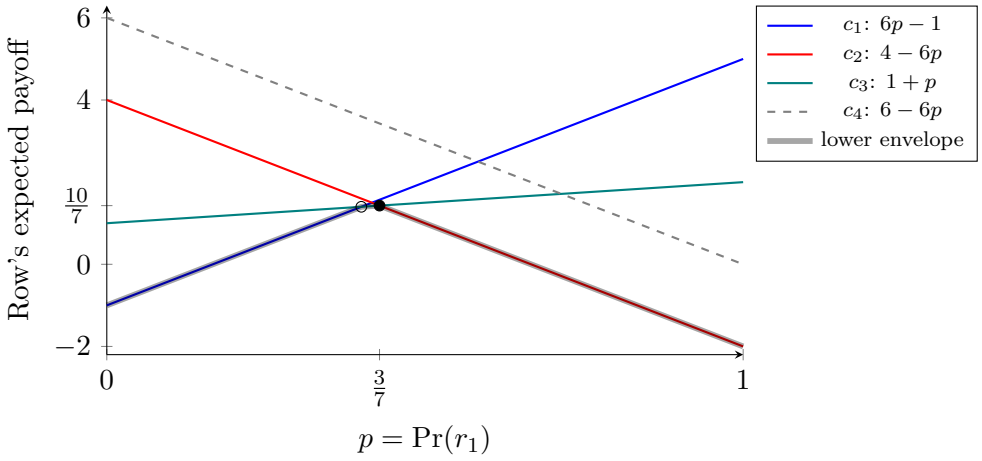
(a) The row minima are -2 and -1, so $\max_i \min_j a_{ij} = -1$. The column maxima are $5, 4, 2, 6$, so $\min_j \max_i a_{ij} = 2$. Since $-1 < 2$, there is no saddle point. Column c_4 is dominated by c_2: $0 > -2$ and $6 > 4$, so Column always pays more with c_4.

Rubric

1 pt: maximin -1 vs. minimax 2. 1 pt: c_4 dominated by c_2 (with direction correct: Column minimizes).

(b) Row's expected payoff against each column when Row plays r_1 with probability p:

$$E_1 = 6p - 1, \qquad E_2 = 4 - 6p, \qquad E_3 = 1 + p, \qquad E_4 = 6 - 6p.$$



The lower envelope (shaded) is $6p - 1$ on $[0, \frac{2}{5}]$ (kink marked by the open circle), $1 + p$ on $[\frac{2}{5}, \frac{3}{7}]$, and $4 - 6p$ on $[\frac{3}{7}, 1]$. It increases up to $p = \frac{3}{7}$ and decreases afterwards, so its maximum is at the intersection of E_3 and E_2:

$$1 + p = 4 - 6p \implies p^* = \tfrac{3}{7}, \qquad v = 1 + \tfrac{3}{7} = \tfrac{10}{7}.$$

Row's optimal strategy is $\boxed{(\frac{3}{7}, \frac{4}{7})}$ and the value is $\boxed{v = \frac{10}{7}}$.

Rubric

2 pts: four lines correct. 2 pts: lower envelope identified correctly, including the short c_3 segment on $[\frac{2}{5}, \frac{3}{7}]$. 2 pts: $p^* = \frac{3}{7}$. 1 pt: $v = \frac{10}{7}$.
Most common error: intersecting c_1 with c_2 (giving $p = \frac{5}{12}$, $v = \frac{3}{2}$) and ignoring c_3. Award 3/7 for that answer if the lines are correct.

(c) At $p^* = \frac{3}{7}$ the columns yield $E_1 = \frac{11}{7}$, $E_2 = E_3 = \frac{10}{7}$, $E_4 = \frac{24}{7}$. An optimal Column strategy must give Row exactly v against p^*, so it can only use c_2 and c_3. Let Column play c_2 with probability y and c_3 with $1 - y$. Row's payoffs:

$$r_1: \ -2y + 2(1 - y) = 2 - 4y, \qquad r_2: \ 4y + (1 - y) = 1 + 3y.$$

Column's strategy is optimal iff both are $\le \frac{10}{7}$: $2-4y \le \frac{10}{7} \iff y \ge \frac{1}{7}$ and $1+3y \le \frac{10}{7} \iff y \le \frac{1}{7}$. So $y = \frac{1}{7}$ and Column's optimal strategy is $\boxed{(0, \frac{1}{7}, \frac{6}{7}, 0)}$; it is **unique**, since the two inequalities pin down y and every other column must receive zero weight.

Rubric

1 pt: identifies c_2, c_3 as the active columns (with reason). 2 pts: $y = \frac{1}{7}$. 2 pts: uniqueness argument (both the support restriction and the two inequalities). Students who find $(0, \frac{1}{7}, \frac{6}{7}, 0)$ by equalizing and assert uniqueness without argument: 3/5.

(d) At $p = \frac{1}{2}$: $E_1 = 2$, $E_2 = 1$, $E_3 = \frac{3}{2}$, $E_4 = 3$. Column best-responds with $\boxed{c_2}$, and Row receives $\boxed{1}$, less than the value $\frac{10}{7}$.

Rubric

1 pt: best response c_2. 1 pt: payoff 1.

Problem 4 Best-Response Correspondences [10 points]

	L	R
U	3, 1	0, 2
D	3, 2	1, 1

$p = \Pr(U)$, $q = \Pr(L)$.

(a) $u_R(U) = 3q$ and $u_R(D) = 3q + (1-q) = 1 + 2q$, so $u_R(D) - u_R(U) = 1 - q \ge 0$ with equality only at $q = 1$:

$$B_R(q) = \begin{cases} \{0\} & 0 \le q < 1, \\ [0,1] & q = 1. \end{cases}$$

Rubric

1 pt: both expected payoffs. 2 pts: correspondence, including the full interval at $q = 1$.

(b) $u_C(L) = p + 2(1-p) = 2 - p$ and $u_C(R) = 2p + (1-p) = 1 + p$, so

$$B_C(p) = \begin{cases} \{1\} & 0 \le p < \frac{1}{2}, \\ [0,1] & p = \frac{1}{2}, \\ \{0\} & \frac{1}{2} < p \le 1. \end{cases}$$

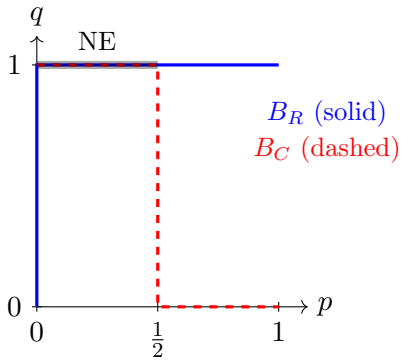
Rubric

1 pt: threshold $p = \frac{1}{2}$. 1 pt: full correspondence.

(c) If $q < 1$ then Row plays D ($p = 0$), but then Column's unique best response is L ($q = 1$), a contradiction. So $q = 1$. With $q = 1$ Row is indifferent, and Column is willing to play L iff $p \le \frac{1}{2}$. The set of Nash equilibria is the segment

$$\boxed{\{(p, q) : 0 \le p \le \tfrac{1}{2},\ q = 1\}},$$

i.e. Column plays L and Row plays U with any probability at most $\frac{1}{2}$. Payoffs: Row 3, Column $2 - p$.



Rubric

1 pt: correct sketch. 3 pts: the full segment. Award 1/3 for finding only (D, L); 2/3 for the segment with a wrong endpoint or an open endpoint at $\frac{1}{2}$.

(d) U is weakly dominated by D ($3 = 3$, $0 < 1$); Column has no weakly dominated strategy. Deleting U leaves Column choosing L, so only $\boxed{(D, L)}$ (i.e. $p = 0$) survives.

Rubric

1 pt: (D, L) with reason.

Problem 5 True or False [8 points]

(a) True. Let $v = a_{rc} = a_{r'c'}$ (all saddle points have the same value). Since (r, c) is a saddle point, $a_{rc} \le a_{rc'}$; since (r', c') is a saddle point, $a_{rc'} \le a_{r'c'}$. Hence $a_{rc'} = v$, which is the minimum of row r and the maximum of column c', so (r, c') is a saddle point.

(b) False. In Problem 4, U is weakly dominated by D, yet $(p, q) = (\frac{1}{2}, 1)$ is a Nash equilibrium in which U is played with probability $\frac{1}{2}$.

(c) True. If a pure strategy in the support earned strictly less than the best pure response, moving its probability to that best response would raise the player's expected payoff, contradicting equilibrium. So all strategies in the support earn the maximal payoff, hence the same payoff.

(d) False. Suppose σ puts weight $\sigma(s) > 0$ on s, which is strictly dominated by τ. Let $\sigma' = \sigma + \sigma(s)(\tau - e_s)$ (move the weight on s to τ). For every opponent strategy y, $u(\sigma', y) - u(\sigma, y) = \sigma(s)\,[u(\tau, y) - u(s, y)] > 0$, so σ is never a best response.

Rubric

Each part: 1 pt for the correct True/False, 1 pt for a valid justification. A correct label with an incorrect or missing justification earns 1/2; an incorrect label earns 0 regardless of the justification.